

I

Introduction: Writing in the Late Age of Print

THE LATE AGE OF PRINT

In a well-known passage in Victor Hugo's *Notre-Dame de Paris, 1482*, the priest Frollo sees in the invention of the printed book an end rather than a beginning:

Opening the window of his cell, he pointed to the immense church of Notre Dame, which, with its twin towers, stone walls, and monstrous cupola forming a black silhouette against the starry sky, resembled an enormous two-headed sphinx seated in the middle of the city. The archdeacon pondered the giant edifice for a few moments in silence, then with a sigh he stretched his right hand toward the printed book that lay open on his table and his left hand toward Notre Dame and turned a sad eye from the book to the church. "Alas!" he said, "This will destroy that" (Hugo, 1967, p. 197).

The priest remarked "Ceci tuera cela": this book will destroy that building. He meant not only that printing and literacy would undermine the authority of the church but also that "human thought ... would change its mode of expression, that the principal idea of each generation would no longer write itself with the same material and in the same way, that the book of stone, so solid and durable, would give place to the book made of paper, yet more solid and durable" (p. 199). The medieval cathedral crowded with statues and stained glass was both a symbol of Christian authority and a repository of medieval knowledge, moral knowledge about the world and the human condition. The cathedral was a library to be read by the religious, who walked through its aisles looking up at the scenes of the Bible, the im-

ages of saints, allegorical figures of virtue and vice, and visions of heaven and hell (Yates, 1966, p. 124). In fact, the printed book did not eradicate the encyclopedia in stone; it did not even eradicate the medieval art of writing by hand. People continued to contemplate their religious tradition in cathedrals, and they continued to communicate with pen and paper for many purposes. However, printing did displace handwriting, in the sense that the printed book became the most highly valued form of writing. Philosophers and scientists of the later Renaissance used the medium of print to refashion the medieval organization and expression of knowledge. As Elizabeth Eisenstein has shown, the printing press has been perhaps the most important tool of the modern scientist (1979, especially pp. 520–574).

Hugo himself lived in the heyday of what we might call “the industrial age of print,” when writers and publishers were taking advantage of mechanized presses to create mass-publication newspapers, magazines, and novels. Hugo’s own popularity in France (like Dickens’ in England) was evidence that such writers were reaching and defining a new audience. Today we are living in the late age of print. Word processing, databases, e-mail, the World Wide Web, and computer graphics are displacing printed communication for various purposes. In the 1980s, the computer and the printed book still seemed to serve different spheres of communication. Computers were well suited to scientific analysis and business data processing and possibly to forms of ephemeral writing, such as memos. Business letters and technical reports might also migrate to the computer, but literary, scholarly, and scientific texts of lasting value would remain in printed form. Now, however, the distinction between lasting texts and pragmatic communication has broken down, and all kinds of communication are being digitized. Major book publishers have for years put their texts in machine-readable form for photocomposition, so that even these texts pass through the computer on their way to the press. It now seems possible that many texts might never be printed, but simply distributed in digital form. The Internet and the World Wide Web have already expanded enormously the uses for digital communication: there are Web sites offering us Greek literature, avant-garde fiction, articles from medical journals, online magazines and newspapers, business materials and advertising for all kinds of products, and both written and visual pornography. Although print remains indispensable, it no longer *seems* indispensable: that is its curious condition in the late age of print. Electronic technology provides a range of new possibilities, whereas the possibilities of print seem to have been played out. As we look up from our computer keyboard to the books on our shelves, we may

be tempted to ask whether “this will destroy that.” The question does not have a definitive answer. What is characteristic of the late age of print is, rather, that we pose the question.

The phrase “late age of print” no doubt makes many readers think of Frederic Jameson and many other neoMarxists who have characterized ours as the age of “late capitalism.” For Jameson (1991), late capitalism does not mean dead capitalism; it means instead a changed system that operates globally through and around traditional governments and cultures (pp. xviii–xxi). Jameson writes, “[w]hat ‘late’ generally conveys is rather the sense that something has changed, that things are different, that we have gone through a transformation of the life world . . . ” (xxi). This is also the best way to think of the late age of print, as a transformation of our social and cultural attitudes toward, and uses of, this familiar technology. Just as late capitalism is still vigorous capitalism, so books and other printed materials in the late age of print are still common and enjoy considerable prestige, especially for the humanities and some of the social sciences. On the other hand, with the rapid decline of socialism, capitalism now seems to have no serious rival as an economic system. The printed book has a rival; indeed, it has had a series of rivals in the visual and electronic media of the 20th century, including film, radio, television, and now digital media. It is these rivalries—especially the latest challenge from digital media—that are now defining how the printed book will function for our culture. Digital media are refashioning the printed book.

Because of the tension between print and digital forms, the idea of the book is changing. For most of us today, the printed book remains the embodiment of text. Both as authors and as readers, we still regard books and journals as the place to locate our most prestigious texts. Few authors today aspire to publish a first novel on the Internet (it is too easy); they still want to be in print. However, the printed book as an ideal has been challenged by poststructuralist and postmodern theorists for decades, and now the computer provides a medium in which that theoretical challenge can be realized in practice. Some groups (scientific researchers along with some in business and government) are already transferring their allegiance from the printed page to the computer screen. They think of the computer as their primary medium, print as a secondary or a specialized one. If our culture as a whole follows their lead, we may come to associate with text the qualities of the computer (flexibility, interactivity, speed of distribution) rather than those of print (stability and authority). As early as 1993, the historian Henri-Jean Martin was willing to claim that that shift in association had already oc-

curred: "Books no longer exercise the power they once did; in the face of the new means of information and communication to which we will have access in the future, books will no longer master our reason and our feelings" (quoted in Chartier, 1995, pp. 13).

It is certainly true that we no longer rely on print exclusively in organizing and presenting scientific and academic knowledge, as we have for the past 5 centuries. The organization of such knowledge now depends on the interplay of printed and electronic forms. The shift to the computer may make writing more flexible, but it also threatens the definitions of good writing and careful reading that have developed in association with the technique of printing. In the heyday of print, we came to regard the written text as an unchanging artifact, a monument to its author and its age. We also tended to magnify the distance between the author and the reader, as the author became a monumental figure, the reader only a visitor in the author's cathedral. In the late age of print, however, we seem more impressed by the impermanence and changeability of text, and digital technology seems to reduce the distance between author and reader by turning the reader into an author herself. Such tensions between monumentality and changeability and between the tendency to magnify the author and to empower the reader have already become part of our current economy of writing.

THE FUTURE OF PRINT

Our culture's ambivalence in the late age of print is reflected in the contradictory predictions made about the future of the printed book and of printed forms in general. The question has been the subject for volumes such as *The Future of the Book* (Nunberg, 1996). The enthusiasts for electronic technology are not ambivalent, and they sometimes predict the end of the book, as Raymond Kurzweil (1999) does:

... [E]lectronic books [of the early 21st century] will have enormous advantages, with pictures that can move and interact with the user, increasingly intelligent search paradigms, simulated environments that the user can enter and explore, and vast quantities of accessible materials. Yet vital to its ability to truly make the paper book obsolete is that the essential qualities of paper and ink will have been fully matched. The book will enter obsolescence, although because of its long history and enormous installed base, it will linger for a couple of decades before reaching antiquity (pp. 297–298).

Sometimes the enthusiasts simply ignore print as they go on to imagine an era of pure and transparent electronic communication, characterized by interactive audio and video or even networked virtual reality. For example, some educators imagine a classroom in which books are replaced by virtual environments:

Applications of virtual reality are being developed in such fields as architecture, medicine, and arcade games ... It is time to see how it could be applied to education and the development of virtual classes in the fullest sense as wraparound environments for learning where students as telepresences can see, hear, touch, and perhaps one day even smell and taste (Tiffin & Rajasingham, 1995, p. 7).

Nor are the critics of electronic writing always ambivalent. Some continue to insist on the division between literary and pragmatic communication—to argue that computers may be used for technical communication and for home entertainment, but that literature will continue to be printed. The novelist E. Annie Proulx claimed in the *New York Times* that “no one is going to read a novel on a twitchy little screen. Ever” (1994, p. A23). Taken literally, this claim is simply wrong. Such conventional novels as *Brave New World* and *Jurassic Park* have been digitized and read (or at least purchased) by an audience of hundreds or a few thousand. Such hypertext fictions as *afternoon* and *Victory Garden*, written exclusively for the twitchy little screen, have also won relatively small, but appreciative audiences. Proulx might be right, if we take her to mean that there will never be a mass audience for verbal fiction in this new medium, and in that case the scientific and literary communities would no longer share a space for publication or a forum for dialogue. Sometimes, too, critics will claim not to be Luddites, but only to be insisting on sensible limits to the computerization of culture: for example, Mark Slouka in *War of the Worlds: Cyberspace and the High-Tech Assault on Reality* (1995). Sometimes a critic will assert that no such limits are possible. In his *Gutenberg Elegies* (1994), Swen Birkerts assumed this fatalistic tone in discussing the eclipse of the printed book: “A change is upon us—nothing could be clearer. The printed word is part of a vestigial order that we are moving away from—by choice and by societal compulsion ... This shift is happening throughout our culture, away from patterns and habits of the printed page and toward a new world distinguished by its reliance on electronic communications” (p. 118). The inevitable was also lamentable: Birkerts spent much of his book, which was, after all, entitled an elegy, la-

menting the passing of the traditional literary culture that he associated with print.

The questions that concern both the enthusiasts and the critics include: What is the nature of the challenge that digital media pose for print? Will digital media replace print? Does the advent of the computer announce a revolution in writing, or is the change less significant? Digital media may challenge traditions of writing at several levels. There is a challenge to print as a technology for delivering alphabetic text and a challenge to the genres and structures that we associate with printed books, newspapers, magazines, and so on. When Proulx complains about reading novels on “twitchy” screens, she assumes that the genre of the novel, which developed in the age of print, will continue to exist in its linear form and denies that computer screens will be the space in which such forms are read. She discounts the challenge that new electronic media might pose to the structure of fiction and nonfiction. In fact, linear forms such as the novel and the essay may or may not flourish in an era of digital media. Writers generally still write with a single, fixed order in mind, but the popularity of the World Wide Web and CD-ROM and DVD is leading some to exploring more fluid structures.

Digital media may also challenge alphabetic writing in any form—in a printed book or on a computer screen. Although printed books, newspapers, and magazines can and do combine graphics with text, new digital media seem often to favor graphics at the expense of text. If in the 1980s, the personal computer was a word processor, it has now become an image processor, which can manipulate and deliver static graphics, animation, and video (as well as audio). Computer graphics are refashioning conventional television and film. The question is whether alphabetic texts can compete effectively with the visual and aural sensorium that surrounds us. And if prose itself is being forced to renegotiate its cultural role, then the printed book is doubly challenged. It is not just that the computer as hypertext can challenge print as a mode of writing; it is also that the printed book is associated so strongly with verbal text. If prose loses its cultural warrant, then who will care about printed books, which are mostly prose? Can printed picture books hope to compete effectively with broadcast television and interactive video? Perhaps printed books will survive as the place for purely verbal texts and for that very reason be pushed to the cultural margin. Prose might in fact have a brighter future, if it could free itself from print technology. In electronic hypertext, for example, prose might combine with audiovisual presentation and perhaps share in the cultural prosperity of the image.

A whole set of cultural questions is connected with the changing status of the word. The importance of verbal literacy in education, the traditional canon, sex and violence on television, censorship in various media—these are all disputes over the appropriate balance between word and image. Much of what American conservatives think of as the “culture wars” is in fact an argument about modes of representation. The number and complexity of these questions suggest that we are at a critical moment in the history of writing. This moment is worth our consideration, no matter how the current tensions between print and digital technology are resolved in the coming decades.

Although it is very difficult to avoid all prediction (in practice, to avoid writing in the future tense), it should nevertheless be possible to resist the impulse to unify—to avoid merging individual predictions into a synthesis that is supposed to represent the one, true future. We should instead treat the predictions of both the enthusiasts and the critics as part of the ambiguous present that constitutes the late age of print. Their predictions reflect the struggles among various cultural factions that are trying to work out the relationship of digital technology to its predecessors. Although we need not try to decide whether the printed book will in fact disappear in 10, 20, or 50 years, we can try to understand the current relationship between print and digital media, which may show us why the future of the printed book seems so uncertain and the future of digital media so bright.

THE OLD AND THE NEW IN DIGITAL WRITING

In this late age of print, digital writing seems both old and new. Although we began in the 1980s by using word processors and electronic photocomposition to improve the production of printed books and typed documents, it has now become clear that we can use the computer to provide a writing surface with conventions different from those of print. A World Wide Web page already differs in some important ways from a conventional printed page. Electronic text takes on shapes that Web designers and other digital authors deem appropriate to the computer’s capacity to structure and present information. In this respect authors and designers are performing the same service for electronic technology that printers performed in the decades following Gutenberg’s invention.

As early as the 1450s and 1460s, Gutenberg and his colleagues were able to achieve the mass production of books without sacrificing quality. Gutenberg’s 42-line Bible does not seem to us today to have been a radical

experiment in a new technology. It is not poorly executed or uncertain in form. The earliest incunabula are already examples of a perfected technique, and there remains little evidence from the period of experimentation that must have preceded the production of these books. Gutenberg's Bible can hardly be distinguished from the work of a good scribe, except perhaps that the spacing and hyphenation are more regular than a scribe could achieve. Because early printers tried to make their books identical to fine manuscripts, they used the same thick letter forms, the same ligatures and abbreviations, and the same layout on the page (Meggs, 1998, p. 63). It took a few generations for printers to realize that they could create a new writing space with thinner letters, fewer abbreviations, and less ink.

The parallel to Gutenberg's period can be overstated, however, for Gutenberg inaugurated the new age of print, rather than the late age of the manuscript. At its invention, the printed book seemed familiar and yet was in many ways new, whereas the computer seems utterly new and revolutionary, when, at least as a writing technology, it still has much in common with its predecessors. Electronic writing is mechanical and precise like printing, organic and evolutionary like handwriting, visually eclectic like hieroglyphics and picture writing. On the other hand, electronic writing is fluid and dynamic to a greater degree than previous technologies. The coming of this new form in fact helps us to understand the choices, the specializations, that the earlier printed book entailed.

Those who tell us that the computer will never replace the printed book point to the physical advantages: the book is portable, inexpensive, and easy to read, whereas the computer is hard to carry and expensive and needs a source of electricity. The computer screen is not as comfortable a reading surface as the page, so that reading for long periods promotes eye-strain. Finally—and this point is always included—you cannot read your computer screen in bed. However, electronic technology continues to evolve. Machines have diminished dramatically in size and in price during the past 40 years, and computer screens are becoming more readable. Some portable computers already have the bulk and weight of notebooks, and it is not hard to imagine one whose screen is as legible as a printed page. In fact, specialized devices styled as electronic books are already commercially available (= > p. 79). We can also envision a system whose flatscreen display is built into the top of a desk or lectern (like those used in the Middle Ages and the Renaissance), where the writer can work directly by applying a light pen instead of typing at a keyboard.

Ease of use is only one measure of a writing technology. The great advantage of the first printed books was not that you could read them in bed. Gutenberg might well have been appalled at the thought of someone taking his beautiful folio-sized Bible to bed. For generations, many important printed books remained imposing volumes that had to be read on bookstands, so that people often read (and wrote) standing up. Mass production by the letterpress did eventually make books cheaper and more plentiful, and this change was crucial. However, the fixity and permanence that printing seemed to give to the written word was just as important in changing the nature of literacy. By contrast, our culture regards digital texts as fluid and multiple structures. If this fluidity seems to offer new possibilities of expression, then writers and readers will put up with some inconveniences to use it.

In place of the static pages of the printed book, the computer can maintain text as a dynamic network of verbal and visual elements. Although writers have been exploiting these dynamic networks for two decades, as long as we are living in the late age of print, electronic writing will seem to be in its infancy. The electronic incunabula include computer-controlled photocomposition, the word processor, the textual database, the electronic bulletin board and mail, and now Web sites. Word processors already demonstrate the flexibility of electronic writing in allowing writers to copy, compare, and discard text with the touch of a few buttons. Change is the rule in the computer, stability the exception, and, as was already realized in the 1980s, it is the rule of change that makes the word processor so useful. On the other hand, most writers have enthusiastically accepted the word processor precisely because it does not challenge their conventional notion of writing. The word processor is an aid for making perfect printed copy: the goal is still ink on paper. Like computer-controlled photocomposition, the word processor is not so much a tool for writing, as it is a tool for typography. (On the interplay between fluidity and fixity in word processing, see Balestri, 1988; Heim, 1987; Mullins, 1988.) The word processor treats text like a scroll, a roll of pages sewn together at the ends, while its visual structures are still typographic. A conventional word processor does not treat the text as a network of verbal ideas. It does not contain a map of the ways in which the text may be read; it does not record or act on the semantic structure of the text. Other forms of electronic writing do all these things, making the text from the writer's point of view a network of verbal elements and from the reader's point of view a texture of possible readings. They permit the reader to share in the dynamic process of writing and to alter the voice of the text.

REFASHIONING THE VOICE OF THE TEXT

Writing in the classical and Western traditions is supposed to have a voice and therefore to speak to its reader. A printed book generally speaks with a single voice and assumes a consistent character, a persona, before its audience. In today's economy of writing, a printed book must do more: it must speak to an economically viable or culturally important group of readers. Our culture has used printing to help define and empower new groups of readers: for example, the middle-class audience for the 19th-century British novel. But this achievement is also a limitation. An author must either write for one of the existing groups or seek to forge a new one, and the task of forging a new readership requires great talent and good luck. Even a new readership, brought together by shared interests in the author's message, must be addressed with consistency. Few publishers would accept a book that combined two vastly different subject matters: say, European history and the marine biology of the Pacific, or Eskimo folklore and the principles of actuarial science. It might even be difficult to publish a book that was part fiction and part non-fiction—not a historical novel, a genre that is popular and has a well-defined audience, but, let us say, a combination of essays and short stories that treat the same historical events. We might say that these hypothetical books lack unity. Yet our definition of textual unity comes from the published work we have read, or more generally, from the current divisions of academic, literary, and scientific disciplines, which themselves both depend on and reinforce the economics of publishing. The material in a book must simply be homogeneous by the standard of some book-buying audience.

This strict requirement of unity and homogeneity is relatively recent. In the Middle Ages, unrelated texts were often bound together, and texts were often added in the available space in a volume years or decades later. Even in the early centuries of printing, it was not unusual to put unrelated works between two covers (= > p. 77). On the other hand, it seems natural to think of any book, written or printed, as a verbal unit. For the book is already a physical unit; its pages are sewn or glued together and then bound into a portable whole. Should not all the words inside proceed from one unifying idea and stand in the same rhetorical relationship to the reader?

Our literate culture is choosing to exploit electronic technology in part to refashion the unified rhetorical voice of the text. Michael Heim (1987) has written, for example, that "Fragments, reused material, the trails and intricate pathways of 'hypertext,' as Ted Nelson terms it, all these advance the disintegration of the centering voice of contemplative thought" (p. 220). An

electronic text may fracture the single voice of the printed text and speak in different registers to different readers. An electronic encyclopedia may address both the educated novice and the expert, just as the same corporate Web site may serve for general public relations, stockholder education, and even sales and marketing. In the ideal, if not in practice, an electronic text can tailor itself to each reader's needs, and the reader can make choices in the very act of reading.

Until recently, the printing press was a classic industrial machine, producing large quantities of identical texts. McLuhan (1972) called printing the first example of the assembly line and mass production (p. 124). Computer-controlled photocomposition has made printing more flexible, helping publishers to produce books more quickly and to target well-defined markets. However, hypertextual writing can go further, because it can change for each reader and with each reading. Authors can exploit the dynamic quality of hypertext to alter the nature of an audience's shared experience in reading. If all the readers of *Bleak House* or *Ulysses* could discuss these works on the assumption that they had all read the same words, no two readers of a hyperfiction can make that assumption. They can only assume that they have traveled in the same textual network. Fixed printed texts can be made into a literary canon in order to promote cultural unity. In the 19th and early 20th centuries, when the canon of literature was often taken as the definition of a liberal education, the goal was to give everyone the experience of reading the same texts—Shakespeare, Milton, Dickens, and so on. This ideal of cultural unity through a shared literary inheritance, which has received so many assaults in the 20th century, must now suffer further by the introduction of new forms of highly individualized writing and reading.

Critics accuse the computer of promoting homogeneity in our society, of producing uniformity through automation, but electronic reading and writing seem to have just the opposite effect. European and North American culture exploited the printing press as a great homogenizer of writing and of the literary audience, whereas that same culture now seems eager to use electronic technologies to differentiate genres and audiences as well as economic markets. In our current world of publication, electronic texts—Web sites, hyperfictions, CD-ROMs and DVDs for entertainment and education—are offered to us as fragmentary and potential texts, each as a network of self-contained units rather than as an organic whole in the tradition of the 19th-century novel or essay. This fragmentation need not imply mere disintegration, however. Elements in the electronic writing space need not

be simply chaotic; they may instead function in a perpetual state of reorganization, forming patterns that are in constant danger of breaking down and recombining. This tension may lead to a definition of effective writing that supplements or replaces our traditional notion of the unity of voice and of analytic argument. What unity there is in an electronic text derives from the perpetually shifting relationship among its verbal elements. What unity there is in the audience for that text comes from the momentary constellation of different economic and cultural “special interests.”

REFASHIONING THE WRITING SPACE

In addition to redefining the voice of the text, our culture is also redefining the visual and conceptual space of writing. Indeed, the spatial metaphor for writing and reading is as culturally powerful now as it has ever been. Cyberspace has become a term for characterizing almost anything to do with the Internet or electronic communication. When we browse the World Wide Web, we think of ourselves as traveling to “visit” the sites, although in fact the servers are delivering pages of information to our computer. The Internet and the Web, CD-ROMS and DVDs, and computer RAM constitute a field for recording, organizing, and presenting texts—a contemporary writing space that refashions the earlier spaces of the papyrus roll, the codex, and the printed book. The continuous flow of words and pages in the book is supplanted in electronic space by abrupt changes of direction and tempo, as the user interacts with a web page or other interface.

Each writing space is a material and visual field, whose properties are determined by a writing technology and the uses to which that technology is put by a culture of readers and writers. A writing space is generated by the interaction of material properties and cultural choices and practices. Moreover, each space depends for its meaning on previous spaces or on contemporary spaces against which it competes. Each fosters a particular understanding both of the act of writing and of the product, the written text, and this understanding expresses itself in writing styles, genres, and literary theories. The writing space is also a space for reading, as Roger Chartier reminds us (Chartier, 1994, p. 2; 1995): communities of readers help to define the properties of the writing space by the demands they place on the text and the technology. For ancient Greece and Rome, the space for writing and reading was the inner surface of a continuous roll, which the writer divided into columns—not because papyrus had to be used this way, but because ancient culture made this choice. The space of the papyrus roll

defined itself in relation to earlier oral forms of communication and to stone or wood inscriptions (= > p. 77). For medieval handwriting and modern printing, the space was the white surface of the page, particularly in a bound volume, which was again a cultural decision of both the Latin and Byzantine Middle Ages. Initially, in late antiquity, the handwritten codex was in competition with the space of the papyrus roll and offered advantages that must have seemed important to contemporary readers. In the 15th century, the printed book defined itself in relation to the manuscript codex that it sought to displace. The space of electronic writing is both the computer screen, where text is displayed, and the electronic memory, in which text is stored. Our culture has chosen to fashion these technologies into a writing space that is animated, visually complex, and malleable in the hands of both writer and reader. In this late age of print, however, writers and readers still often conceive of text as located in the space of a printed book, and they conceive of the electronic writing space as a refashioning of the older space of print.

Because writing is such a highly valued individual act and cultural practice, the writing space itself is a potent metaphor. In the act of writing, the writer externalizes his or her thoughts. The writer enters into a reflective and reflexive relationship with the written page, a relationship in which thoughts are bodied forth. Writing, even writing on a computer screen, is a material practice, and it becomes difficult for a culture to decide where thinking ends and the materiality of writing begins, where the mind ends and the writing space begins. With any technique of writing—on stone or clay, on papyrus or paper, and on the computer screen—the writer may come to regard the mind itself as a writing space. The behavior of the writing space becomes a metaphor for the human mind as well as for human social interaction. Such cultural metaphors are in general redefinitions of earlier metaphors, so that in examining the history of writing, and in particular electronic writing today, we should always ask: How does this writing space refashion its predecessor? How does it claim to improve on print's ability to make our thoughts visible and to constitute the lines of communication for our society?

2

Writing as Technology

It is not difficult to recognize the printing press, the typewriter, and the Linotype machine as technologies. These industrial-age means of book production were complex and to a degree self-activating or self-directing. The mechanization of writing began in the 15th century with the letter press itself, which was the first text “processor,” the first technology of writing to duplicate words en masse. (On the development of printing, see Steinberg, 1959.) In replacing the scribe who formed letters one at a time, the press registered several pages of text onto a large sheet of paper with each impersonal pull. As McLuhan recognized in *The Gutenberg Galaxy* (1972), “the invention of typography ... provid[ed] the first uniformly repeatable commodity, the first assembly line, and the first mass-production” (p. 124). Printing had the additional virtue that it could produce books that were nearly identical to the best manuscripts: the press rivaled handwriting in quality while far surpassing it in quantity. Elizabeth Eisenstein (1983) noted that “[t]he absence of any apparent change in product was combined with a complete change in methods of production, giving rise to the paradoxical combination of seeming continuity with radical change” (p. 20). This paradox made the mechanization of the word easier to accept, and scholars in the 15th century immediately saw the advantages of this new technology. Gradually, over several generations, printing did change the visual character of the written page, making the writing space technically cleaner and clearer. The book had taken on a different and more modern look by the late 18th century. In the 19th and 20th, steam and electric presses, and automatic typesetting brought further mechanization, further distancing the human controller from the printing process and defining an industrial age of print. The computer in turn adds new flexibility to the rapidity and efficiency of printing. The computer’s capacity to adjust the text to each user’s

needs, which is uncharacteristic of the classic industrial machine, derives from the unmechanical materials of electronic technology. The computer's central processor itself contains no gears or indeed any moving parts above the level of electrons, and even the mechanical components of a computer system, such as disk drives and printers, are characterized by rapid movement and fine control. The digital computer suggests a new definition of the machine, as a complex interrelation of logical as well as physical parts that, unlike the steam engine and the dynamo, processes information rather than producing power. If the printing press was the classic writing machine, the computer constitutes a technology of writing beyond mechanization, a postindustrial form of writing.

The medieval manuscript and ancient papyrus roll in turn represent technologies of writing before mechanization. These preindustrial technologies also required devices—pen and paper or parchment in the Middle Ages or reed pen and papyrus in the ancient world. Working from such raw materials as rags, animal skins, or plants to produce a finished book certainly required considerable technical knowledge. Still, the manuscripts were produced at the relatively slow pace of the scribe's hand, not the insistent rhythm of the machine, and we can see on each page the variations in size and shape of letters that indicate direct human production. The development of mechanical printing and now writing by computer has affected our view of these previous writing techniques. A medieval manuscript, rubricated and bound in leather, would perhaps have struck a Greek in Plato's time as a complicated and ingenious device. When we admire the manuscript as art, however, what we admire is the apparent negation of technology, the fact that the codex is not a printed book and therefore not the product of a machine.

There are good historical (as well as etymological) reasons, however, for broadening the definition of technology to include skills as well as machines. The Greek root of "technology" is *techne*, and for the Greeks a *techne* could be an art or a craft, "a set of rules, system or method of making or doing, whether of the useful arts, or of the fine arts" (Liddell & Scott, 1973, p. 1,785). In his dialogue the *Phaedrus*, Plato calls the alphabet itself a *techne*. He would also have called the ancient book composed of ink on papyrus a *techne*; even Homeric epic poetry was a *techne*, as was Greek tragedy. All the ancient arts and crafts had this in common: that the craftsman must develop a skill, a technical state of mind in using tools and materials. Ancient and modern writing are technologies in the sense that they are methods for arranging verbal ideas in a visual space. The writer always needs a surface on

which to make his or her marks and a tool with which to make them, and these materials become part of the contemporary definition of writing. Writing with quill and parchment is a different skill from writing with a printing press, which in turn differs from writing with a computer. However, all writing entails method, the intention of the writer to arrange verbal ideas in a space for later examination by a reader. In *Orality and Literacy* (1982) Walter Ong argued that writing is “interiorized” and that the process of interiorization makes it difficult for us to recognize writing itself as a technology (pp. 81–82).

Our technical relationship to the writing space is always with us as readers and writers. Literacy is, among other things, the realization that language can have a visual as well as an aural dimension, that one’s words can be recorded and shown to others who are not present, perhaps not even alive, at the time of the recording. Literate people know that words can be placed in a visual space and have continued existence in that space. They always know this. Even when they are speaking, they know that their words can be written down. Students of culture as uncongenial as Walter Ong and Jacques Derrida have insisted that writing is a constant presence in our mental life. Cultures with a long tradition of literacy develop a standard literary language, so that men and women reveal their literacy when they are speaking as well as writing. Illiterates are denied access to that language, while those educated in the tradition tend to speak in a combination of colloquial and literary terms. They speak, as they write, in a variety of styles and levels, and they often structure their speech as they do their writing, talking in sentences and even paragraphs. They write in their mind as well as on paper or at a keyboard; indeed, they are writing whenever they think or verbalize in that methodical way characterized by writing.

There may be many such literacies at any given historical moment. A cultural elite may succeed in defining a high literacy, whose *techne* excludes various minorities. North American culture, for example, was relatively successful in maintaining such a high literacy until perhaps the Second World War: this *techne* was embodied in the literary canon that some traditionalists still wish to affirm as the basis for education. Minorities, however, may develop their own *technes*, their own internalized literacies. Spoken language can itself be a *techne*, for it can require method—most obviously in the varieties of oral poetry and storytelling that have been practiced for thousands of years and are still important today in much of the developing world. (On Homeric oral poetry, see Lord, 1968; Thomas, 1992, pp. 29–51. On the many other varieties of oral poetry, see Finnegan, 1977.) The oral

poet applies method to language in order to create verse forms and story structures, although in this case the structures must be able to be appreciated by listening rather than by reading. The oral poet is a writer, who writes exclusively in the minds of his or her audience.

Despite its apparent immediacy, however, oral poetry is no more natural than writing, just as writing with pen and paper is no more natural, no less technological, than writing on a computer screen. Although the computer is a more complicated and fragile device than a pen, we could not isolate ourselves from technology by reverting to older methods of writing. The production of today's pens and paper also require a sophisticated manufacturing process. Without electricity, industrial organization, and networks of transportation and distribution, we could not provide ourselves with adequate supplies of these simple writing materials. It is not the complexity of the devices that matters so much as the technical or literate frame of mind. Writing technologies are never external agents that invade and occupy the minds of their users. These technologies are natural or naturalized only in the sense that they are constituted by the interaction of physical materials and human practices. No technology, not even the apparently autonomous computer, can ever function as a writing space in the absence of human writers and readers. And what Walter Ong characterized as the literate mind is simply another name for the collective decisions shared among writers and readers about how to exploit their materials in order to communicate.

WRITING TECHNOLOGIES AND MATERIAL CULTURE

As Christina Haas (1996) puts it: "Writing is situated in the material world in a number of ways. It always occurs in a material setting, employs material tools, and results in material artifacts" (p. 4). Roger Chartier has made the argument repeatedly in his work on the history of the book:

"Whether they are in manuscript or in print, books are objects whose forms, if they cannot impose the sense of the texts that they bear, at least command the uses that can invest them and the appropriations to which they are susceptible. Works and discourses exist only when they become physical realities ... This means that ... keen attention should be paid to the technical, visual, and physical devices that organize the reading of writing when writing becomes a book" (Chartier, 1994, pp. viii-ix).

What Chartier says of books and manuscripts is certainly also true of digital technology (see also Duguid, 1996). The materiality of writing matters,

as much for electronic writing as for earlier forms. Electronic writing still requires our physical interactions with terrestrial materials—with the keyboard, the mouse, and the computer screen. Despite the dreams of cyberpunk authors and virtual reality enthusiasts, repetitive stress injuries should serve as a reminder that computers are physical entities. Electronic writing is always involved in material culture and in contemporary economics. Computers are tangible objects with aesthetic and social significance, and digital devices and software are objects of important economic exchange. A single software program, an operating system, has made one man perhaps the richest private citizen in the world, but more important has had a significant impact on economic activity throughout the developed world.

Electronic writing may also be virtual, yet all previous writing technologies were virtual as well, in the sense that they invited writers and readers to participate in an abstract space of signs. This virtual or conceptual writing space forms a continuum with the material space of writing: it is both a reflection of contemporary materials and techniques and an expression of our culture's ambitions for its writing. Just as new digital media refashion the material conditions of print and handwriting, so the computer's virtuality refashions the writing space of the printed book and the manuscript. The electronic writing space may seem to be severed from the material world in a way that the space of print was not. Enthusiasts for the electronic writing space often promote as revolutionary its dynamic and apparently autonomous character—the fact that Web pages and multimedia applications can act as well as react, redefining themselves in ways that neither the author nor the reader can necessarily predict in advance. Yet those actions and reactions are grounded in the physics and computer science of the technology and are the result of specific interventions on the part of human writers (or programmers) and readers (or users). Our literate culture is simply using the new tools provided by digital technology to reconfigure the relationship between the material practices of writing and the ideal of writing that these practices express. It remains as true for the computer as for earlier technologies that the materiality of writing “must be acknowledged to fully appreciate the nature of literate acts” (Haas, 1996, p. 3). (For a discussion of materiality and theories of language and writing in the 20th century, see Johanna Drucker's *The Visible Word*, 1994, particularly pp. 9–47.)

Chartier reminds us that writing never exists only in the abstract:

“Readers and hearers, in point of fact, are never confronted with abstract or ideal texts detached from all materiality; they manipulate or perceive objects and forms whose structures and modalities govern their reading (or their

hearing), thus the possible comprehension of the text read (or heard)" (Chartier, 1994, p. 3).

Each technology of writing involves different materials or different ways of deploying the writing materials, and these differences are significant. This is not to suggest, however, that the material conditions of writing determine in an exclusive fashion how a literate culture will read and write its texts. The relationship of the material techniques and the nature of writing—what Haas (1996) calls the Technology Question (pp. 3–23)—is far subtler and more interesting (see also Grusin, 1996). The very materiality of writing binds writing firmly to human practices and therefore to cultural choices. The technical and the cultural dimensions of writing are so intimately related that it is not useful to try to separate them: together they constitute writing as a technology. The technology of ancient writing is not only the papyrus, the ink, and the techniques of making book rolls; it is also the styles and genres of ancient writing and the social and political practices of ancient rhetoric. The technology of modern writing includes not only the techniques of printing, but also the practices of modern science and bureaucracy and the economic and social consequences of print literacy. If personal computers and palmtops, browsers and word processors, are part of our contemporary technology of writing, so are the uses to which we put this hardware and software. So too is the rhetoric of revolution or disaster that enthusiasts and critics weave around the digital hardware and software.

It is not a question of seeing writing as an external technological force that influences or changes cultural practice; instead, writing is always a part of culture. It is probably best to understand all technologies in this way: technologies do not determine the course of culture or society, because they are not separate agents that can act on culture from the outside. Yet the rhetoric of technological determinism remains common today. Popular writers often seem to suggest that technologies, especially digital information technologies, are agents in this sense. The World Wide Web, virtual reality, or computers are said to revolutionize our society, our economy, and even the way we think. More substantial writers such as McLuhan and Ong can also sound like technological determinists: McLuhan when in *Understanding Media* (1964) he explores media as "extensions of man", and Ong when he claims in *Orality and Literacy* (1982) that writing restructures consciousness. The very title of Elizabeth Eisenstein's massive book on the age of print, *The Printing Press as an Agent of Change* (1979), suggests that she too is guilty of technological determinism. On the other hand, it is possible to understand print technology is an agent of change without insisting that

it works in isolation or in opposition to other aspects of culture. Even the Marxist sociologist of media, Raymond Williams, warned us to avoid economic or social determinism, just as we should avoid technological determinism (1975, pp. 130).

Individuals and whole cultures do mold techniques and devices to their own purposes, but the material properties of such techniques and devices also impose limitations on their possible uses. There are many things we cannot do with contemporary computers, even things that some would obviously like to do. For example, artificial intelligence has not provided us machines with the capacity to write stories or create fully autonomous graphic worlds, though many technophiles would like to have them. It also seems that a technique or device can render certain social constructions easier and or even possible. The hardware and software of the Internet made it possible to construct the World Wide Web. Our culture chose to turn the Web into a carnival of commercial and self-promotional Web sites, but that carnival would not be possible without the underlying hardware and software protocols. Yet even this way of putting it seems to create a false dichotomy between devices (“hard” technology) and social uses. Even in their brief period of development, the hardware and software of the Internet have clearly changed in response to social and economic pressures, and these social and economic pressures respond in turn to each new technical possibility. The open architecture of the World Wide Web allowed individuals to create sites and add them to the Web without the approval of any authority. This openness led to erotic Web sites, which in turn led to calls in the United States to block such sites because children might visit them. Some politicians wanted to censor these sites by law, and the ensuing struggle has also led to the development of software filters to allow parents to block certain sites. Wherever we start in such a chain of cause and effect, we can identify an interaction between technical qualities and social constructions—an interaction so intimate that it is hard to see where the technical ends and the social begins. When I speak of a technology of writing throughout this book, I will in general mean not just the hard technology, but the sum of the technical and social interactions that constitute a writing system. It is the contemporary technology of digital writing in this broad sense that we are trying to characterize.

In this characterization, the material properties remain significant. The properties of handwriting, of print, or of digital writing do each seem to favor certain kinds of expression and to prejudice others. The printed book favors linear writing; the computer makes associative linking easier. A technology,

as it has been culturally constructed, can predispose us toward a particular definition of “natural” writing. Thus, if a writer chooses to display fixed, linear prose on a computer screen, she is working “against the grain” of the technology, just as Lawrence Sterne in the 18th or the dadaists and other avant-gardes in the 20th century have worked against the medium of print to create highly associative prose. The very fact that such writers exploit the technology in other than its “natural” mode gives their work its significance. Even in these cases, however, we do not have to argue that linear writing is inherently natural to print. The linear character of print is the outcome of the constant interaction between the properties of the printed book and the decisions that Western authors and readers have made about how to exploit those properties. In other words, we can reject the attempt to isolate the technical from the social aspect of technologies of writing, but we can still examine how these techniques and materials have been deployed by writers and readers at various cultural moments. Linear writing is appropriate to print technology both because the printed page readily accommodates linear text and because our culture expects that printed prose should be linear. Other styles may be appropriate to digital technology, where software systems like the World Wide Web are both expressions of and influences on our changing attitudes toward writing.

ECONOMIES OF WRITING

Each culture and each period has had its own complex economy of writing, a dynamic relationship among materials, techniques, genres, and cultural attitudes and uses. The earliest economies flourished in Mesopotamia and in Egypt, where complex word-syllable scripts were recorded on stone, clay, or papyrus (see Gaur, 1984; Gelb, 1963; Jensen, 1969; Sampson, 1985). The ancient Greeks and Romans borrowed both the materials and the elements of their writing economy: the Greek alphabet was taken from the Phoenicians, and papyrus from Egypt served as the chief writing material for the Greeks and the Romans. The ancient book was a roll, consisting of sheets of papyrus glued together at the ends. The paged book or codex, which came into use in the 2nd and 3rd centuries A.D., opened a range of new possibilities for writing, by offering a space both more varied and more accessible than that of the roll (See Chartier, 1995, pp. 18–20; Reynolds & Wilson, 1978, pp. 30–32) (= > p. 78). The codex was put to more sophisticated use in the Western European economy of writing. In the Middle Ages parchment provided a more durable and more attractive writing surface than pa-

pyrus, which in any case became hard to obtain from Egypt. Paper was introduced from the Far East as a cheaper replacement for parchment in the later Middle Ages. Although not as tough as parchment, paper made from rags could be produced in greater quantities and could therefore supply a growing demand for reading materials (see Febvre & Martin, 1971, pp. 39–60. See also Gaur, 1984, pp. 44–47). With these new materials, medieval scribes slowly refashioned the writing space they had inherited from the ancients; they employed word division, punctuation, rubrication (decorated initial letters), headings, and letter styles to organize the text visually on the page. They began to insert critical notes and glosses into the margins of the text, sometimes in several layers. In some scholarly medieval codices, the page became a web of text and interpretation, tradition and innovation. Finally, the invention of printing in the 15th century initiated the modern economy of writing with its highly organized and standardized space. During each of these periods, there have been numerous secondary technologies as well: wax tablets, chalkboards, typewriters, stenographs, dictaphones, and so on—each fulfilling needs that could not be easily met by the dominant technology (see, for example, Rouse & Rouse, 1989).

When in the history of writing a new technology appears, it may supplement an established technology or replace it. Papyrus was replaced in the Middle Ages by parchment and paper. In the late 19th and early 20th centuries, the typewriter replaced handwriting for business communications. At the same time American society was beginning to accept women into the work force so that the change created secretarial jobs for young women, who took the place of male clerks (see Zuboff, *In the Age of the Smart Machine*, 1988, pp. 115–116). Now the word processor has replaced the typewriter. Whenever a dominant technology is challenged, there may be a major refashioning of the culture's writing space. The three dominant technologies since ancient times, the papyrus roll, the codex, and the printed book, each participated in the fashioning of a rather different writing space. When the codex replaced the roll, it refashioned the writing space from the still relatively oral space of ancient culture to the progressively more visual and less oral space of medieval writing. When the printed book supplanted and marginalized the codex, the writing space took on the qualities of linearity, replicability, and fixity that we have associated with the printed book. Electronic and digital technology are helping to refashion the writing space again. In the late age of print, this refashioning is not complete, and we are now experiencing the tensions and inconsistencies that come from attempts either to reconcile the two spaces of print and digital technology or definitively to replace the one with the other.

In its role as a great refashioner, electronic writing is reintroducing characteristics that have belonged to a variety of marginal techniques of the past. Electronic writing shares with the wax tablet or chalkboard the quality of rapid and easy change. It shares with the typewriter its keyboard (at least at present), its method of discrete selection of alphabetic elements, and its mechanical uniformity; with improvements in speech recognition, electronic writing systems can function like a tape recorder in taking input through a microphone. The computer can serve as a copier, a note pad, a calendar, or a teletype machine. In fact, it is hard to think of a marginal technology in the history of writing that the computer cannot imitate, just as it is hard to think of a dominant technology whose elements the computer does not borrow and reinterpret. Electronic writing may therefore participate in the restructuring of our whole economy of writing. Of greatest importance, however, is the way in which our literate culture is using the computer to refashion the printed book, which, as the most recent dominant technology, is the one most open to challenge.

REMEDIATION

In about the 8th century B.C., the Greeks began to refashion the space of oral mythology and heroic legend into the more precise and linear space of the papyrus roll (and stone or wooden inscription), a process that, according to Eric Havelock (1982), lasted hundreds of years. In late antiquity the shift from papyrus roll to codex refashioned the space again, making more effective use of the two-dimensional surface to deploy text. In Western Europe the shift from handwritten codex to printed book was another such refashioning, and the shift to electronic writing is yet another. We might call each such shift a "remediation," in the sense that a newer medium takes the place of an older one, borrowing and reorganizing the characteristics of writing in the older medium and reforming its cultural space. Writing on papyrus remediated oral communication by involving the eye as well as the ear and so giving the words a different claim to reality. The other shifts too blatantly or subtly changed the terms on which we as readers approach the text and its mode of representing the world. Remediation involves both homage and rivalry, for the new medium imitates some features of the older medium, but also makes an implicit or explicit claim to improve on the older one.

Remediation is a process of cultural competition between or among technologies. For centuries, the Greeks and Romans conceived of their technology of alphabetic writing on papyrus roll in a dialectic tension with the oral

tradition that writing only partly replaced. Ancient prose, even philosophy and history, was often highly rhetorical, as if the writing were still trying to imitate and improve on oral presentation. The shift from codex to papyrus roll was less problematic, with the result that the codex remediated the roll almost out of existence in a few centuries. In the Renaissance the printed book remediated the manuscript by appearing to provide the same visual space as the manuscript with the added benefits of mass production (= > p. 78). Over centuries, however, the printed book was a significant refashioning that defined a space in which fixity and accuracy were more highly prized than perhaps ever before.

Digital technology is turning out to be one of the more traumatic remediations in the history of Western writing. One reason is that digital technology changes the “look and feel” of writing and reading. A printed book could and did at first look like a manuscript, its appearance changing gradually over several decades. Chartier (1995) argues that the current shift from print to electronic technology, which he calls a revolution, entails a change greater than the one from manuscript to print:

“Our current revolution is obviously more extensive than Gutenberg’s. It modifies not only the technology for reproduction of the text, but even the materiality of the object that communicates the text to readers.... The substitution of screen for codex is a far more radical transformation because it changes methods of organization, structure, consultation, even the appearance of the written word.... The present revolution has only one precedent in the West: the substitution of the codex for the *volumen*—of the book composed of quires for the book in the form of a roll—during the first centuries of the Christian era” (pp. 15, 18).

Yet, until the 1980s, it was not apparent to most readers and writers that the computer was a writing technology at all. Before the advent of word processing on personal computers, our literate culture regarded computers as “number-crunching” tools for engineers or as electronic filing cabinets for bureaucratic data. In the past two decades, however, computers have been recognized not only as writing technologies, but as media for popular entertainment and expression, which we are using to refashion visual as well as verbal communication.

Remediation is not limited to technologies of writing. Richard Grusin and I have examined the ways in which new visual media, such as computer graphics, virtual reality, and the World Wide Web, define themselves by borrowing from, paying homage to, critiquing, and refashioning their predecessors, principally television, film, photography, and painting (Bolter &

Grusin, 1999). Computer games remediate film by styling themselves as “interactive movies”; virtual reality remediates film as well as perspective painting; digital photography remediates the analog photograph. The World Wide Web absorbs and refashions almost every previous visual and textual medium, including television, film, radio, and print. Furthermore, older media can remediate newer ones within the same media economy. Today, the traditional cinema is attempting to maintain its status by employing computer graphics in conventional linear films. And television is making such extensive use of new media that TV screens often look like pages from the World Wide Web. Remediation is a characteristic process not only for contemporary media, but for all visual media at least since the Renaissance with its invention of linear-perspective painting. Each medium seems to follow this pattern of borrowing and refashioning other media, and rivalry as well as homage seems always to be at work.

Furthermore, since the Renaissance, our culture has had two apparently contradictory expectations for its visual media. In one sense the goal of representation has been transparent presentation. The medium is supposed to function as a window through which the viewer can see the objects represented. That was in fact exactly how the artist and writer Leon Battista Alberti characterized linear-perspective painting in his treatise “On Painting”: as a window on the world (Alberti, 1972, p. 55). Western artists and audiences have generally treated perspective painting, photography, film, and now virtual reality and three-dimensional computer graphics as transparent media. On the other hand, artists and their audiences do not always want the medium to disappear; they often want to be made conscious of and even surrounded by media. Instead of transparency, they strive for *hypermediacy*, an intense awareness of and even reveling in the medium. Contemporary television, for example, is often hypermediated, although it can sometimes function as a transparent medium as well. Among new digital media, the World Wide Web is most often characterized by hypermediacy. The same medium can strive for transparency in one case and hypermediacy in another, and in general today we swing back and forth between a desire for transparent contact with the ostensibly real (unmediated) world and a fascination with the possibilities that media offer us. Because the number of old and new, analog and digital, media available to us today is very great, we live in an environment that is conducive to hypermediacy. Yet the desire for a transparent medium remains strong.

What all media and media forms have in common for our culture is the promise of immediacy. Transparent media promise to disappear and leave us

in contact with the unmediated world, although it is a promise that they can never entirely fulfill. Hypermediated media give up the attempt to present a world beyond themselves; instead, they offer themselves as immediate experiences. When one medium sets out to remediate another, it does so by claiming to do a better job. It can claim to be better at transparency. For example, virtual reality promises to be the ultimate transparent medium, better than painting or photography, because the viewer in virtual reality can actually step into the world viewed. Or the medium can promise a more elaborate hypermediacy, as World Wide Web sites do in combining painting, photography, graphic design, film, audio, and video into a sort of popular *Gesamtkunstwerk*. In either case the new medium is trying to convince us that it offers greater immediacy than its predecessors. Because our culture today is saturated with media, claims of greater immediacy are constantly being made, as new and older media vie for our attention.

The remediations of writing technologies are like those of visual media, particularly in the case of digital media where words and images combine and interact so freely (= > p. 47). Furthermore, throughout the 20th century, print has engaged in contests of remediation with photography, film, and television. All these visual technologies are still remediating print, while digital technologies are working their remediations too. The best way to understand electronic writing today is to see it as the remediation of printed text, with its claim to refashioning the presentation and status of alphabetic writing itself. The qualities that distinguish electronic writing from print, flexibility and interactivity, become the bases of the enthusiasts' claim that the computer can improve on the printed book. For the enthusiasts, these qualities can make the experience of reading sometimes more transparently real, sometimes more hypermediated, but always more immediate. As we shall see, to say that electronic writing is flexible and interactive is to say that it is hypertextual.